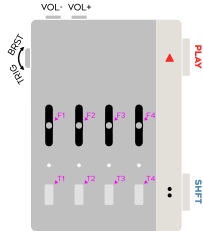


Wakes v0.4.8 - Pg.1 - Welcome



- Hey! I'm Adara. Welcome to **Wakes**, a custom firmware for **Teenage Engineering's** unreleased "SP-1" stem player. **Wakes** combines two Eurorack modules into one device: **Plaits**, a macro-synthesizer, and **Marbles**, a playful random sequencer. The two modules were created by **Émilie Gillet**, otherwise known as **Mutable Instruments**.

- First and foremost, you should know how to turn the device on or off. Holding down **SHIFT** for 1.5 seconds will show a brief animation before the device will turn on, going to its first page, **Plaits**.

- While turned on, holding **SHIFT** for 3 seconds will turn the device off.

- While plugged in, **Wakes** has a standby mode.

- While on, any input while **SHIFT** is held down will interrupt the shutdown behaviour, allowing you to work inside **SHIFT** pages.

- Even if interrupted, holding down **SHIFT** for 25 seconds will cause the face LEDs to flash quickly, notifying you that the SP-1 is about to perform an emergency shutdown in 5 seconds if you don't let **SHIFT** go. Release and return to holding **SHIFT** to get back to what you were doing. This is intended as an emergency backstop.

- At any point, even when off, tapping **SHIFT** for a short moment will bring up a battery meter.

- While using **Wakes**, you'll be bouncing between **Plaits** to design sounds, and **Marbles** to sequence it.

- As you bounce back and forth, the face LED's will show you what's going on through a system of **glyphs**.

- Faders **F1-F4** have a catch-up mechanism with the actual digital values, so moving them around and swapping pages doesn't create unexpected behaviours. For clarity, follow the LED's to know where the set values reside.

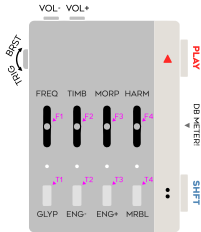
- Each module has a **SHIFT** page containing advanced parameters and options that open up avenues for experimenting.

- Double-tapping **SHIFT** enters a deep tinkering page called **SETTINGS**. There are two **SETTINGS** pages: One for **Plaits**, one for **Marbles**.

- Legend: ○ - Lit LED ● - Unlit LED



Wakes v0.4.8 - Pg.2 - Plaits - BASE



- **Wakes'** first page is **Plaits**: A synthesizer containing a multitude of sound engines representing different sound design techniques, with parameters attached to explorative macros. A **macro-synthesizer**.

- **Plaits** has four main faders: **FREQ**, **TIMB**, **MORP** and **HARM**. Each of these controls several components of the selected synthesis engine.

- Plaits has a built-in **lowpass-gate** and **amplifier**, controlled by an internally-routed **decay envelope**.

- For now, you can read about what the main parameters do for each engine in **commonly available Plaits guides**.

- While in Plaits, the four side LED's will show a dB meter. Each LED represents 6dB of dynamic range.

- Generally speaking, **FREQ** will control the overall pitch of the sound. **TIMB** sweeps the spectral content from dark/sparse to bright/dense. **MORP** explores lateral timbral variations. **HARM** controls the frequency spread/overall balance of the spectrum.

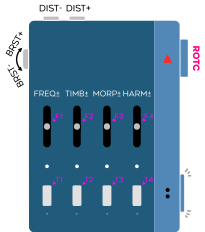
- The rocker switch plays notes, triggering the decay envelope on the filter/amp. Press it down to send a single trigger, pull it up to send a burst.

- Engines are known by **glyphs**. Pressing **GLYP** will flash the glyph representing the selected engine. **ENG+** for the next one, **ENG-** to go back.

- **PLAY** engages **Marbles'** clock, which is patched by default to trigger Plaits and give it musical notes.

- Press **MRBL** to go to **Marbles**.

Plaits - SHFT



- Hold down **SHFT** to go to **Plaits SHFT**.

- Each fader here represents a **modulation depth control** for incoming control signals. By default, Plaits' **decay envelope** is assigned to all of the four parameters. Use the faders to modulate Plaits' macros! Fun!

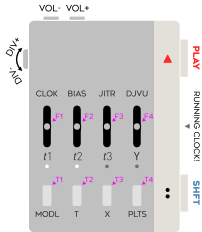
- The rocker switch selects a rhythmic subdivision for **BRST's** retriggers.

- **DIST-** and **DIST+** control the depth of a master softclipper.

- Holding down any of **T1-T4** for 2 seconds **Unpatches** Marbles modulators from that parameter.

- Holding down **PLAY** for 3 seconds **Rips Out Plaits' Cables**, returning it to its default configuration.

Wakes v0.4.8 - Pg.3 - Marbles - t Side (left)



• Marbles has two sides: **t** and **X**. The glyphs will show you which side you are on: ○○○● for **t** and ●○○○ for **X**. Each side has 3 outputs: **t1-t3**, **X1-X3**. An extra output **Y** is offered. It's weird. :)

• **t** is a random gate generator. It emulates a player following a stable metronome. **CLOK** determines the BPM (default: 120). **BIAS** determines the distribution of gate signals depending on the selected model using **MODL**. **JTR** changes the player's accuracy. stable > chaotic.

• **DJVU** controls the probability of reusing played patterns. At center, it is a locked loop. Moving down adds more unlocked randomness. Up from centre, memorized notes are randomized instead. **DJLN** sets loop length.

• **t2** is always a stable clock output. Pressing **MODL** swaps between a list of distribution models for **t1** and **t3**:

○○○○ - coin toss: gates are evenly distributed.

●○○○ - clusters: random pairs of clock subdivisions appear at each side.

○○○○ - drums: Steps through drum pattern tables containing rhythms.

●○○○ - independent coin: each side tosses its own. **t1** and **t3** can coincide.

○○○○ - divider: **BIAS** selects from fixed subdivision pairs. 8:1 > 1:1 > 1:8.

○○○○ - three states: a three-way choice per tick: **t1**, **t3** or neither.

• **DIV-/DIV+** divide/multiply the clock.

• Press **T3** to go to the **X** side of Marbles.

• Press **PLTS** to go back to **Plaits**.

• Hold down **SHFT** to go to **t SHFT**.

t SHFT

• **T1-T4** select **Plaits** modulation targets for **t1-t3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○○○-TIMB, ●○○○-MORP, ●○○○-HARM, ●○○○-LEVL, ●○○○-TRIG. For **Y**, ●○○○ routes to **FREQ**, contextually.

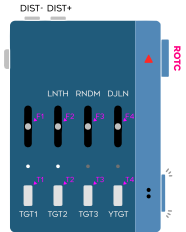
• **LNTH** determines the length of on-states for **t1-t3** gate signals.

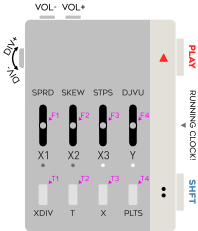
• **RNDM** determines gate length randomization.

• Holding down any of **T1-T4** for 2 seconds **Unpatches** any **Plaits** destinations.

• Holding **PLAY** for 3 seconds **Rips Out all of Marbles' Cables**.

Marbles - t SHFT



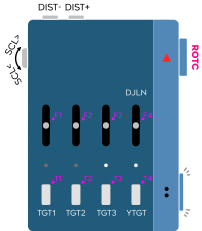


- **X** is a random modulation generator. Its three outputs, **X1-X3**, are clocked by the outputs of Marbles' **t** side, and generate random control signals that can vary from Sample & Hold style steps to smooth, meandering modulations. It can modulate Plaits' parameters, or quantize it to play musical intervals, creating melodic sequences, as well as enabling many other ideas.
- The fourth output, **Y**, shows up on this page as well. It is a smooth, meandering full-range random source that's clocked at 1/8th the rate of **t2**. **Y** never obeys **DJVU**, she's wild like that.
- **SPRD** spreads the concentration of control signals around the central point determined by **BIAS**.

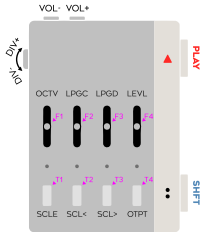
- **SKEW** skews the signals **X1-X3** up or down in average intensity.
- **STPS** smooths or tightens the vertical steppiness of the control signals. Below-centre, the signals have random intensities. Above-centre, it **quantizes X to a musical scale**.
- **DJVU** here is the same as on the **t** side. In fact, both sections are controlled by the same memory.
- Press **XDIV** to choose the intensity diversity between the **X** outputs: **identical**, **bump** (**X2** is strongest) or **tilt** (**X1** is weakest, **X3** is strongest).
- Press **T2** to go back to **t**.
- Press **T4** to go back to **Plaits**.
- Hold down **SHFT** to go to **X SHFT**.

X SHFT

- **T1-T4** select Plaits **modulation targets** for **X1-X3**, or for **Y**. The mod targets are: ●●●●-none, ○●●●-FM, ●○●●-TIMB, ●●●○-MORP, ●●●○-HARM, ●○●○-FREQ, ●●●○-LEVL. **FREQ** is the default input for musical tuning. **FM** is full-range.
- **DJLN**, like in **t**, controls the length of the **DJVU** sequence. **DJVU** and **DJLN** are the same across the two sides.
- **SCL</SCL>** select from a range of scales to **quantize** into, when **STPS** is above the middle point. View base available scales in the Marbles **SETTINGS** guide page.
- Holding down any of **T1-T4** for 2 seconds **Unpatches** any Plaits destinations.



Wakes v0.4.8 - Pg.5 - Plaits - SETTINGS



- Reach Plaits' **SETTINGS** page by double-tapping **SHIFT** while in the Plaits module. Single-tapping **SHIFT** will return you to where you were.

- **OCTV**'s default is its maximum value. When not maxed, it locks Plaits' available frequencies to 2 octaves. Choose the octave range to play by moving the fader up and down. Maximize it to go full-range again.

- **LPGC** lets you control the color of the **low-pass gate**, which is composed of a filter and amplifier. Lower values mix in more of the low-pass filter, making the **internal decay envelope** and **LEVL** sound duller in tone.

- **LPGD** lets you change the length of the decay envelope. Higher values set longer lengths, meaning triggered notes decay for longer when played. Any self-patching from the unpatched **SHIFT** faders will also be longer. Shorter values sound snappier and quieter overall.

- **LEVL** changes the base offset applied to Plaits' filter and amp. Moving it around fades the sound in and out. The color of this fade is affected by **LPGC**. Setting **LEVL** to maximum will effectively **disable them**, allowing Plaits to **drone**. Some Plaits engines will react beautifully to modulating **LEVL**. It's available as a mod destination for **t** and **X**. It's the default mod destination affected by Plaits' **decay envelope**.

- **Wakes** includes a smart Plaits **quantizer** feature. It lets you noodle around with notes on a scale by moving **FREQ**, without Marbles' modulation. It is ******-off** by default. Activate it by pressing **SCL>**. Deactivate it by pressing **SCL<** until the glyphs show ******-off** again. Once activated, it first disables any Marbles routings going into **FREQ**. Then, it takes over **FREQ** and quantizes it. Combine it with **OCTV** to lock your available notes to a short range. **SCL** flashes the currently-active **quantizer scale**. Scales are the same as the ones available for Marbles, with glyphs shown in Marbles' **SETTINGS** manual page.

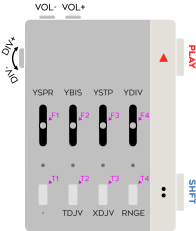
- Plaits features an extra output, **AUX**, containing additional content from each engine. Wakes gives you **OTPT** on this page to select from different output configurations: ******-OUT**, ******-AUX**, ******-OUT+AUX**, and the Wakes-unique ******-OUT×AUX**.

Plaits Tips

- Plaits is designed to be really breakable. Experiment with fast **BRST**, quick Marbles modulations, lots of mod targets. Just go crazy with it, and see what wilderness you end up in. **Ripping Out The Cables** will take you back home.

- **FM** is a legacy Plaits input containing exp-TZFM for Plaits. It's used for atonal modulations.

- You can **TRIG** while **LEVL** is open ;)



- Reach Marbles' **SETTINGS** page by double-tapping **SHIFT** while in the Marbles module. Single-tapping **SHIFT** will return you to where you were.
- **YSPR** lets you control Y's spread, same way you do for X.
- **YBIS** lets you control Y's randomization skew towards higher or lower signal strengths.
- **YSTP** lets you control its overall steppiness. Lower values are more sinusoidal. Central values are linear, triangular. Higher values are slewed squares, up to an even sample & hold.
- **YDIV** lets you choose at which subdivision of the clock Y is clocked. This lets you accelerate it to higher values if you want.

- **TDJV** and **XDJV** toggle **DJUV** for the torX sides on or off. **DJUV** is on for both by default. If you toggle it off for one of the sides, that side won't be affected by **DJUV**'s looping.
- **RNGE** sets the control signal strength range for X and Y. The options are: **0000** - 0-20%, **0001** - 0-50%, **0002** - $\pm 50\%$, **0003** - **INTELLIGENT**. **INTELLIGENT** is the default. It's a Wakes-specific upside of combining the two modules in a closed environment: It decides your voltage range selection for you, per-output, automatically, and with the target Plaits engine in mind. That said, if you still want to play around with unexpected results, the other modes remain available.

- There are 6 default scales Marbles offers for quantizing the X outputs. When you press **SCL<** or **SCL>** on Marbles' **SHIFT** page, their glyphs will flash on the face LEDs as you cycle through the list. Here they are: **0000**-major, **0001**-minor, **0002**-pentatonic, **0003**-pelog, **0004**-bhairav, **0005**-shri. If you choose one of these scales, then move **STPS** above-center, then send one of the X outputs into **FREQ**, Marbles will tell Plaits to play notes on that scale.

Marbles Tips

- You can combine multiple Marbles outputs into one Plaits destination. Using **Unpatch** disconnects only that specific output from its destination.